

Pragnya Devi Golrilla

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Experience Summary

Results-driven Data Engineer with 5+ years of experience building cloud-native data platforms on **GCP** and designing scalable ETL pipelines using **SQL, Python, PySpark, Dataproc, Dataflow, Pub/Sub, Cloud Data Fusion, and Databricks**. Specialized in **BigQuery performance tuning, cost optimization, and Delta Lake on GCS**. Skilled at creating real-time streaming solutions, data models powering advanced analytics with **Looker, Tableau, and Power BI**. Adept at working with cross-functional teams to deliver **high-impact workflows**, while committed to continuous learning and innovation.

Work Experience

Data Engineer | American Express

Jan 2024 – May 2025

- Developed & optimized advanced SQL queries on 1M+ row datasets, improving query performance by 40%
- Built **GCP** based pipelines on **Databricks** to support batch processing 30GB/day and reducing latency time by 25%
- Engineered a real-time data pipeline using **Spark Structured Streaming & Pub/Sub**, simulating 100K+ events/hour.
- Designed **partitioned star schema models in BigQuery**, accelerating dashboard query speed by 3 times.
- Leveraged **Gen AI** to automate data cleaning, accelerate SQL query and ETL code generation, and implement intelligent pipeline monitoring, enhancing efficiency and reducing errors.
- Maintained versioned **Delta Lake tables on GCS** and used Python (pandas) for orchestration, logging, error handling, and data validation, ensuring reliable storage and robust pipeline monitoring.
- Created dbt models for transformation logic and testing, improving data quality and pipeline transparency.

Data Engineer | Value Momentum

Dec 2020 – July 2023

- Built scalable ETL pipelines with **Dataproc & PySpark**, processing 100GB+ of structured, semi-structured data daily.
- Orchestrated 100+ batch pipelines with Cloud composer, enhancing system reliability & reducing failure rates.
- Automated 15+ ingestion workflows with Cloud Functions, Airflow and GCS, cutting manual interventions by 90%.
- Streamlined BigQuery SQL with **partitioned & clustered tables**, boosting reporting query speed by 35% overall.
- Reduced system downtime by 20% by implementing partitioned tables for time-series snapshot reporting.
- Decreased migration effort by 45+ hours by aligning schema transformations for cloud scalability and performance.
- Implemented scalable Python frameworks for data validation and reconciliation, improving accuracy by 30%.
- Created Slowly Changing Dimension Type 2 models and dimensional schemas to support historical trend reporting.
- Collaborated with cross-functional teams in Agile sprints, Composer/BigQuery pipeline delivery and deployments.

Data Engineer | GenY Medium

June 2020 – Nov 2020

- Implemented 5+ data workflows using Python and SQL to consolidate multi-source marketing data for analysis.
- Developed interactive Power BI dashboards with real-time performance metrics, reducing reporting time by 50%
- Collaborated with data science teams to integrate **ML model outputs and AI** to boost campaign CTR by 15%

Data Engineer Intern | GenY Medium

Dec 2019 – May 2020

- Optimized SQL queries for exploratory data analysis and KPI metric generation for business reporting.
- Supported the migration of on-prem data to GCS, gaining hands-on experience with cloud data management.

Data Engineer Intern | Digithon

May 2019 – July 2019

- Assisted in data preparation and reporting using SQL, Excel, and visualization tools for internal business insights.
- Conducted initial data analysis to identify trends and anomalies, supporting weekly business review presentations.

Technical Skills

- **Programming & Scripting:** Python, SQL, PySpark, R, Scala
- **Cloud:** GCP (BigQuery, Cloud DataFusion, Pub/Sub, DataFlow), AWS (Glue, Redshift, EMR, kinesis, Athena),
- **Data Engineering:** Apache Spark, Databricks, Hadoop, Kafka, Airflow, dbt, Snowflake, BigQuery, Redshift.
- **Visualization:** Looker, Tableau, Power BI, QuickSight, Advanced Excel
- **DevOps & Collaboration:** Docker, Kubernetes, Git, custom logging, Jira, Confluence, Kanban

Education

- **M.S. in Business Analytics and Information Systems | University of South Florida, Tampa** Aug 2023 – May 2025
- **B.Tech in Computer Science Engineering | JNTU–H, INDIA** Aug 2016 – Sept 2020

Certifications

- *Databricks Certified Data Engineer Associate, 2025*
- *Google Cloud Certified Professional Data Engineer – 2024*